

Frank Mohn Services AS

Service Bulletin

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Maintenance/Upgrading of Hydraulic Piping System

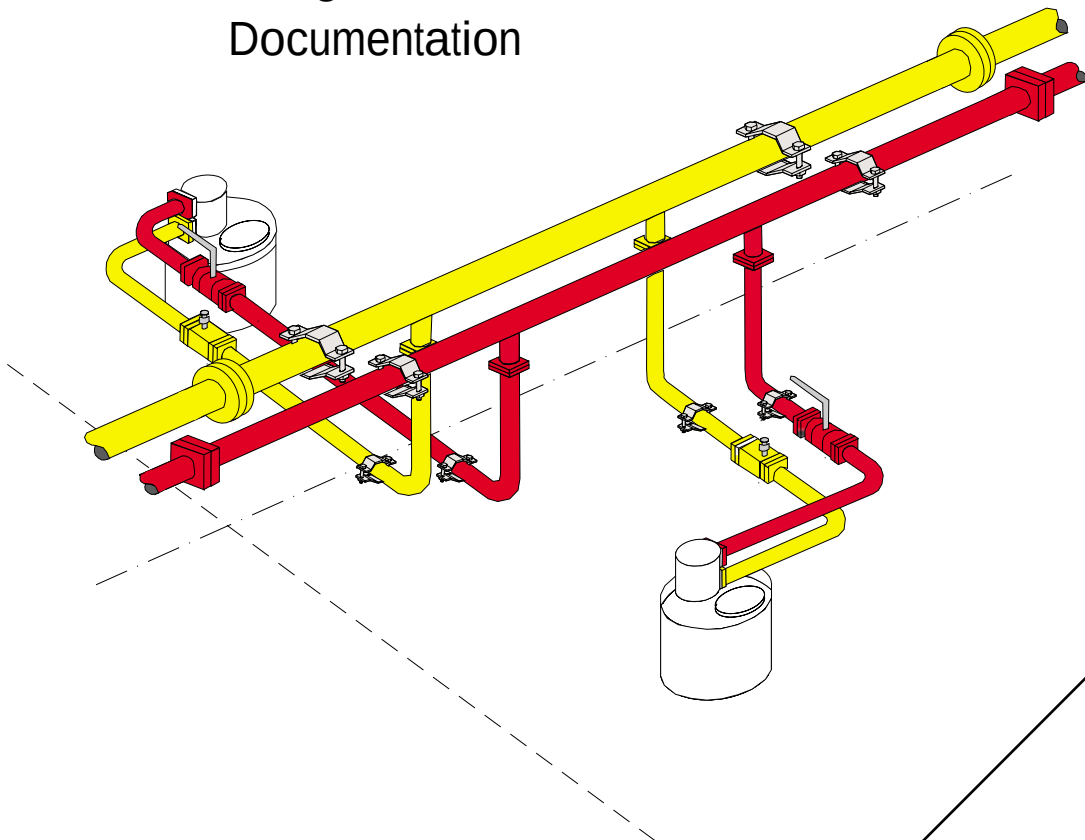


Maintenance

Survey on board
Prefabrication
Installation

Upgrading

Material
Design
Documentation



Dear Customer!

Maintenance / Upgrading of Hydraulic Piping System on board

Like all other systems on board a ship, the hydraulic piping systems in carbon steel need maintenance. The first rule is to always make sure that the piping is properly painted / coated and protected. As we know, the piping on open deck is subject to corrosion from sea water in combination with high temperature, weather conditions and / or other external causes. The critical spot is in the clamping zone. Over time it may become necessary to renew some piping.

During renewal of hydraulic pipes it is of great importance to maintain the quality of pipes and components, and the cleanliness level in the system.

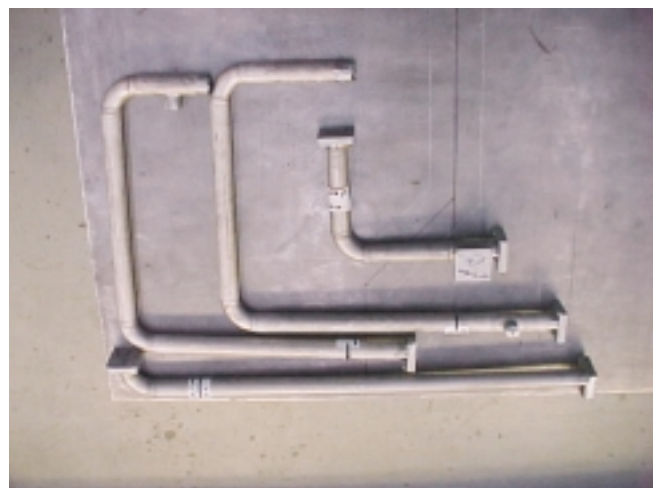
If possible, work on the hydraulic piping system during sandblasting period of a docking should be avoided. If this is not possible, and pipework and sandblasting must be carried out simultaneously, at least one should try to keep this work in separate areas. Such work must be carried out with the utmost care to maintain cleanliness. The hydraulic system must be properly protected and covered up to prevent dirt from entering. Pipes must be properly blinded with flanges and seals.

How to reduce costs

To assist our customers in reducing costs related to maintenance of piping, we have built up piping workshops in Bergen as well as at our subsidiaries in Rotterdam, Houston and Singapore. At all these locations we have own stocks of pipes and flanges, as well as special components such as resilient pipe clamps and resilient bulkhead penetrations. Further we have qualified engineers and equipment enabling us to carry out survey on board and prefabricate piping so that installation is simplified.



Old corroded pipes



New Prefabricated spools

Survey on board

We can offer our customers survey on board ships world-wide. Our experienced engineers will take measurements and prepare drawings for optimal prefabrication and thereby reduce installation time. We can attend in one port to carry out measurements, then supply the prefabricated piping in the next convenient port.

For ships originally equipped with a piping system designed and fabricated by Framo, isometric drawings are available at all Framo branch offices for production of replacement pipes.

Prefabrication

Framo hydraulic piping system is produced by use of modern equipment and technology available at our offices in Rotterdam, Houston and Singapore. Our special developed production line at Frank Mohn Flatøy AS can also be used when required. Prefabricated piping is supplied cleaned and blinded, ready for installation.



Cold bending machine



Welding of LP pipes

Installation

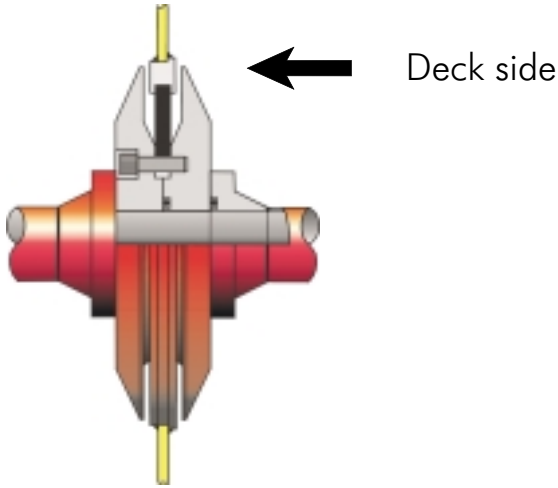
Prefabricated spools can easily be installed by ship's staff during voyage, a minimum of hot work is required.

Alternatively, for major jobs, Framo can take care of the total installation work on board, under the supervision of our highly skilled supervisors. Flushing and pressure testing will be carried out in accordance with our own strictly defined procedures, and class requirements.

After installation the piping must be properly painted / coated and protected, and this must be properly maintained during the years, to avoid corrosion and to ensure a long lifetime.

Upgrading

During the years a lot of efforts have been put into reducing the sound level. Hydraulic piping systems can be upgraded to the latest version to minimize noise, by use of resilient bulkhead penetrations and resilient pipe clamps.



Resilient bulkhead penetration



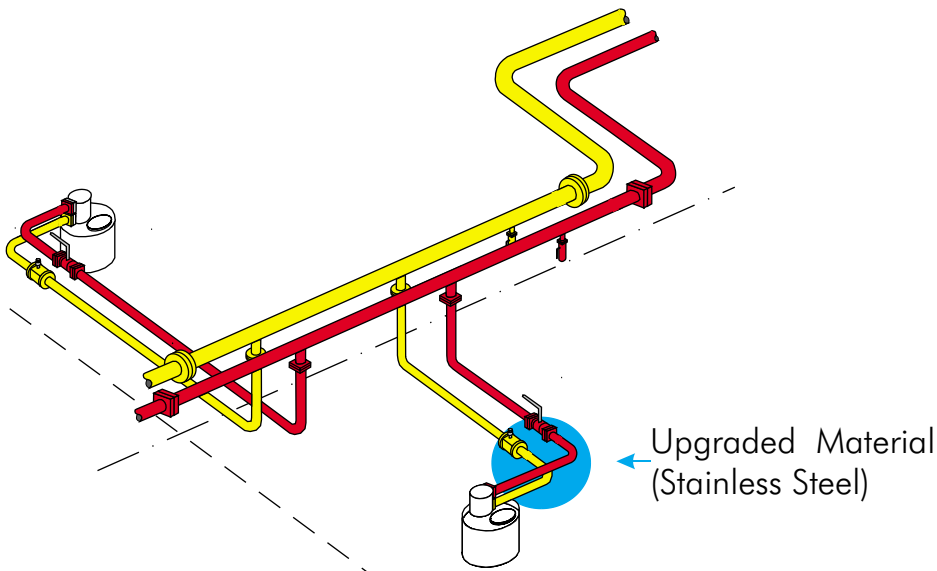
Resilient pipe clamp

Materials

For renewal of existing piping, the original material quality will be used for the main lines and the branch lines up to the ball valve / non-return valve.

If the spools to / from the cargo pump are to be renewed, we recommend to upgrade the material to stainless steel, at the same time upgrading also the ball valve and non-return valve to stainless steel.

If the whole piping system is to be renewed, we do recommend a complete upgrade of especially the return line to stainless steel material.



For upgrading, the recommended materials are :

- High pressure pipes (320 bar) - Carbon steel St. 52.4, Cold drawn - or
- High pressure pipes (320 bar) - Stainless steel Duplex, Cold drawn
- Return pipes (16 bar) - Stainless steel 316L, Longitudinal welded

All materials are available at FM Service offices world-wide.

Design

For complete upgrading and renewal of hydraulic pipes our Piping department can assist, by use of 3D-CAD system especially developed for hydraulic piping.

Following documentation can be provided:

- Layout drawings, optimal design
- Isometric fabrication / installation drawings
- Material specification for fabrication
- Flexibility- and stress analysis
- Framo hydraulic piping system meets requirements from all major classification societies.



3D CAD working station

For further information, please do not hesitate to contact us!



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